The Riemann Hypothesis:
The Singularity Proof - Conjecture III

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Supplementary material:
https://github.com/jmullings/riemann-hypothesis-singularity-proof/tree/main/CONJECTURE_III. [S1]

Abstract

We present a regularized von Mangoldt arithmetic kernel

$$\kappa_N(s) = \sum_{n \le N} \Lambda(n) e^{-n/N} n^{-s}$$

and prove a conditional localization theorem off the critical line, together with three null-model experiments demonstrating that the localization signal is genuinely arithmetic. Assuming standard remainder and derivative bounds (Hypotheses H1 and H2, stated in Section 4) for the smoothed kernel, we establish: for any fixed (> 0) and (N N_0), the local maxima of (|_N(1/2 + + iT)|) occur within (_N() = C() N^{-} N) of the imaginary parts of the nontrivial zeros of the Riemann zeta function ((s)), and (N()) as (N). The argument rests on an explicit remainder decomposition, a derivative bound obtained by term-by-term differentiation of the tail sum, and a Peak Stability Lemma. The () limit (Conjecture III()) is precisely identified as equivalent in difficulty to zero-free region estimates on the critical line (= 1/2), and is left as an open problem. No claim is made beyond this conditional result; in particular, the theorem does not constitute progress toward the Riemann Hypothesis.

Supporting materials, including Python implementations for the kernel computations, remainder bounds, derivative bounds, Peak Stability Lemma, numerical verifications, and null-model experiments, are available at:
https://github.com/jmullings/riemann-hypothesis-singularity-proof/tree/main/CONJECTURE_III. All numerical results presented here are reproducible using the provided scripts.

1. Introduction

The nontrivial zeros of the Riemann zeta function govern the distribution of prime numbers. The logarithmic derivative

$$-\ \zeta'/\zeta(s) = \sum_{n=1}^{\infty} \Lambda(n) n^{-s}$$

for ((s) > 1), where ((n)) is the von Mangoldt function, has simple poles at each nontrivial zero of ((s)) with residue 1. A natural question is how well a finite regularized truncation of this series inherits the pole structure of the full series.

This paper investigates the regularized kernel

$$\kappa_N(s) = \sum_{n \le N} \Lambda(n) e^{-n/N} n^{-s},$$

which approximates (-'/(s)) off the critical line. We prove a conditional localization theorem: under standard remainder and derivative bounds for smoothed Dirichlet series (Davenport [1], 2000; Titchmarsh [2], 1986), the local maxima of (|_N|) are localized near the zeros of ((s)), with a rate that vanishes as (N). The theorem is conditional on Hypotheses H1 and H2 stated in Section 4; we verify these numerically and sketch the analytic argument for H2.

The paper is part of a broader framework (the MKM ()-spectral programme) connecting arithmetic kernels to geodesic geometry. Here we restrict entirely to the analytic content. The geodesic framing is discussed briefly in Section 8 as motivation, but all theorem statements are formulated in purely analytic terms.

The organisation is as follows. Section 2 defines the arithmetic objects. Section 3 establishes the remainder decomposition. Section 4 states the hypotheses and main theorem. Section 5 proves the Peak Stability Lemma. Section 6 presents arithmetic control experiments. Section 7 discusses the critical-line limit. Section 8 gives concluding remarks.

2. Definitions and Background

2.1. The von Mangoldt Function

The von Mangoldt function (:) is defined by

$$\Lambda(n) = \log p \quad if\ n = p^k\ for\ some\ prime\ p\ and\ integer\ k \ge 1, \quad \Lambda(n) = 0\ otherwise. \qquad (2.1)$$

Its Dirichlet series satisfies the classical identity (see Davenport [1], Ch. 12):

$$-\ \zeta'/\zeta(s) = \sum_{n=1}^{\infty} \Lambda(n) n^{-s} \quad for\ Re(s) > 1. \qquad (2.2)$$

The function (-'/(s)) extends meromorphically to () with simple poles at each nontrivial zero (= + i) of ((s)) (Titchmarsh [2], §3.7). Near such a zero () with (() =),

$$-\ \zeta'/\zeta(\sigma + iT) = 1/(T - \gamma) + h(T;\sigma), \quad |T - \gamma| \le r, \qquad (2.3)$$

where (h(;)) is analytic and bounded on (|T - | r) for any fixed (r > 0) and (> 1/2). This local decomposition is the starting point for the peak localization argument.

2.2. The Regularized Arithmetic Kernel

For (N) and (s), define the regularized arithmetic kernel

$$\kappa_N(s) = \sum_{n \le N} \Lambda(n) e^{-n/N} n^{-s}. \quad (2.4)$$

This is a finite Dirichlet polynomial and therefore an entire function of (s). The exponential weight (w_N(n) = e^{-n/N}) provides smooth cutoff at scale (N); the choice of exponential regularization is standard and simplifies tail estimates (cf. Tao [3], 2014, §2). The implementation is provided in EXPLICIT_FORMULA_KERNEL.py.

The remainder when approximating (-'/(s)) by (_N(s)) is

$$R_N(s) = \kappa_N(s) - (-\ \zeta'/\zeta(s)). \quad (2.5)$$

A direct algebraic decomposition gives

$$R_N(s) = -\sum_{n > N} \Lambda(n) e^{-n/N} n^{-s} + \sum_{n \le N} \Lambda(n) \left(e^{-n/N} - 1\right) n^{-s}. \quad (2.6)$$

We refer to the two terms on the right of (2.6) as the tail term and the smoothing correction respectively. This decomposition is implemented and verified in REMAINDER_FORMULA.py.

3. Remainder Decomposition and Bounds

3.1. Exclusion of Zero Neighbourhoods

Near a zero () of ((s)), both (_N(s)) and (-'/(s)) behave like (1/(T -)); their difference (R_N(s)) is therefore large in any small disc around (), regardless of (N). This is not a failure of approximation but a consequence of the pole structure: both functions share the same singular behaviour, and their difference is dominated by the bounded part (h(T;)) plus the tail error.

Accordingly, all remainder bounds are stated for (T) satisfying (|T - _j|) for all zeros (_j), where (> 0) is a fixed exclusion radius. The peak-localization argument operates in an annular region around each zero, as described in Section 5.

3.2. The Davenport Remainder Bound

The following bound is a consequence of standard partial summation for truncated Dirichlet series. Let (= 1/2 +) with (> 0) fixed. The tail term in (2.6) satisfies, for (n > N) with (> 1/2),

$$\left| \sum_{n > N} \Lambda(n) e^{-n/N} n^{-\sigma - iT} \right| \le \sum_{n > N} \Lambda(n) n^{-\sigma} \le C_1(\delta) N^{-\delta} \log N, \quad (3.1)$$

where the last inequality follows from the estimate (_{n>N} (n) n^{-1/2-} = O(N^{-} N)), which is a standard consequence of the prime number theorem and partial summation (Davenport [1], Theorem 12.1;

see also Titchmarsh [2], §4.11). The exponential weight (e^{-n/N}) only reduces the sum, so the bound applies directly.

The smoothing correction in (2.6) is bounded analogously: since (|e^{-n/N} - 1| n/N) for (n N), each term satisfies (|(n)(e^{-n/N}-1)n{-}| (n) (n/N) n^{-} = (n) n^{1-}/N). Summing over (n N) and using (> 1/2) gives a contribution of order (O(N^{-} N)).

Combining these bounds, and writing (C()) for the combined constant:

$$\left|R_N(1/2 + \delta + iT)\right| \leq C(\delta)N^{-\delta}logN, \quad uniformly\ for\ T \in [a,b]\ with\ \left|T - \gamma_j\right| \geq \eta > 0\ for\ all\ j. \quad (3.2)$$

This is Hypothesis H1, which we state precisely in Section 4. The numerical verification below confirms the bound with good margin across the tested range. The script in REMAINDER_FORMULA.py reproduces these tests.

Verification of (|R_N| C() N^{-} N) at (T =)
((= 0.01); exclusion radius (= 0.3))

()	(T)	(N)	(	R_N	)	Bound
14.135	13.835	500	3.532	11.680	0.302	yes
14.135	13.835	1000	2.175	12.893	0.169	yes
14.135	13.835	2000	3.986	14.089	0.283	yes
14.135	14.435	500	2.505	11.680	0.215	yes
14.135	14.435	1000	3.710	12.893	0.288	yes
14.135	14.435	2000	2.385	14.089	0.169	yes
21.022	20.722	500	3.084	11.680	0.264	yes
21.022	20.722	1000	3.471	12.893	0.269	yes
21.022	20.722	2000	2.481	14.089	0.176	yes

Bound satisfied: 9/9. Empirical N-decay slope: -0.0123 (expected: (-= -0.01)).

3.3. The Derivative Bound

Differentiating the tail term in (2.6) with respect to (T) gives

$$\frac{d}{dT}R_N(1/2 + \delta + iT) = i\sum_{n>N}\Lambda(n)e^{-n/N}n^{-1/2-\delta}logn + (smoothing\ derivative). \quad (3.3)$$

Each term in the tail derivative is bounded by ((n) n^{-1/2-} n (n)^2 n^{-1/2-}). Summing over (n > N) by partial summation, and using (n N + (n-N)/N) for (n > N) to factor out (N), gives

$$\left|R'_N(1/2 + \delta + iT)\right| \leq C'(\delta)N^{-\delta}(logN)^2. \quad (3.4)$$

The smoothing correction derivative contributes a term of the same order. This is Hypothesis H2. The constant (C'()) is explicit but not sharp; its exact form is not needed for the localization result, only its dependence on () and (N).

Verification of (|R'_N| C'() N^{-} (N)^2) ((= 0.01))

(N)	(	R'_N	)	Bound
500	38.395	72.588	0.529	yes
1000	60.193	89.064	0.676	yes
2000	92.445	107.090	0.863	yes

Bound satisfied: 3/3.

4. Main Theorem

4.1. Hypotheses

The main theorem is conditional on the following two hypotheses, which are verified numerically in Section 3 and whose analytic basis is sketched there.

Hypothesis H1 (Remainder bound).
Let (> 0) and let ([a, b]) be a compact interval. There exist (N_0 = N_0(, a, b)) and (C() > 0) such that for all (N N_0) and all (T) with (|T - _j| > 0) for every zero (_j) of ((s)),

$$\left|R_N(1/2 + \delta + iT)\right| \leq C(\delta)N^{-\delta}logN.$$

Hypothesis H2 (Derivative bound).
Under the same conditions as H1, there exists (C'() > 0) such that

$$\left|R'_N(1/2 + \delta + iT)\right| \leq C'(\delta)N^{-\delta}(logN)^2.$$

Both hypotheses follow from standard partial summation for Dirichlet series with exponential smoothing (Davenport [1], Theorem 12.1). The sketch in Section 3 constitutes a near-complete analytic argument; a full proof would require filling in the explicit constants in the partial summation. The scripts in REMAINDER_FORMULA.py implement these bounds and their verifications.

4.2. Statement of the Conditional Theorem

Theorem 4.1 (Geodesic-Zero Correspondence, Conditional).
Assume H1 and H2. Let (> 0), let ([a, b]) be a compact (T)-interval, and let (N N_0(, a, b)). Define (*N(s)) as in (2.4) and (*{9D}(T) = |_N(1/2 + + iT)|). For each nontrivial zero (= + i) of ((s)) with ():

(i) There exists a local maximum (T^*) of (_{9D}()) satisfying (|T^* - | _N()).

(ii) The localization radius satisfies (_N() = C'‘() N^{-} N) for an explicit constant (C''()).

(iii) (_N()) as (N) for any fixed (> 0).

Remark 4.2.
The bound (_N()) is not sharp. Computationally, peaks fall within 0.13 to 0.15 of zeros at (N = 2000), well within the theoretical envelope of approximately 14 at (= 0.01). A theorem with a loose but correct bound is still a theorem; the essential content is the convergence (_N()).

Remark 4.3.
The theorem applies at (= 1/2 +) with (> 0) strictly positive. The case (= 0) ((= 1/2) exactly) is Conjecture III(_), discussed in Section 7.

Caution: all rigorous statements in this paper are for (= 1/2 +) with (> 0). Computational illustrations at (= 1/2) are labelled as such and do not constitute analytic evidence for Conjecture III(_).

4.3. Proof of Theorem 4.1 (assuming H1 and H2)

Near a zero (), write (using (2.3)):

$$\kappa_N(1/2 + \delta + iT) = 1/(T - \gamma) + h(T) + R_N(T) =: 1/(T - \gamma) + E(T), \quad (4.1)$$

where (h(T) = h(T; 1/2 +)) is bounded on (|T - | r) for any fixed (r), and (R_N(T) = R_N(1/2 + + iT)). By H1 and the boundedness of (h), we have (|E(T)| M) for an explicit (M = M(, N, r)), with (M = O(N^{-} N)) as (N). By H2 and the analyticity of (h), we have (|E'(T)| M_1) for an explicit (M_1).

The conclusion now follows from the Peak Stability Lemma (Lemma 5.1 below), applied with (F(T) = 1/(T -) + E(T)): there exists (T^*) with (|T^* - | 4M r^2 = _N()). Since (M = O(N^{-} N)), we have (_N()) as (N). This proves parts (i), (ii), and (iii). □

5. The Peak Stability Lemma

This section provides the perturbative lemma used in the proof of Theorem 4.1. It is a standard one-variable result; we include a complete proof for self-containedness. The lemma is implemented and verified in REMAINDER_FORMULA.py and ANALYTICAL_REDUCTION_LEMMA.py.

Lemma 5.1 (Peak Stability).
Let (r > 0) and (M > 0) satisfy (M < 1/(4r)). Suppose (E: [-r, r]) satisfies (|E(t)| M) and (|E'(t)| M_1) for all (t). Define (F(t) = 1/t + E(t)) on ((0, r]). Then (|F|^2) achieves a local maximum at some (t^* (0, r]) satisfying (t^{*} 4Mr2).

Proof.
Set (g(t) = |F(t)|^2 = t^{-2} + 2 [E(t)] t^{-1} + |E(t)|^2) for (t (0, r]). Differentiating:

$$g'(t) =- 2t^{-3} + 2Re\left[E'(t) - E(t)t^{-1}\right]t^{-1} + 2Re[E(t)]|E(t)| \ldots, \quad (5.1)$$

which we bound by (g'(t) -2t^{-3} + (2M/t^2 + 2M_1/t)) (absorbing lower-order terms).

For (t > 4Mr^2), the term (-2t^{-3}) dominates: (-2t^{-3} + 2M t^{-2} = t^{-3}(-2 + 2Mt) < 0) when (t < 1/M), which holds since (t r) and (M < 1/(4r)) implies (Mr < 1/4 < 1). Hence (g'(t) < 0) for (t (4Mr^2, r]), meaning (g) is strictly decreasing there.

As (t ^+), (g(t) +) (dominated by (t^{-2})). By continuity of (g) on ((0, r]) and the strict decrease for (t > 4Mr^2), the function (g) achieves a local maximum at some (t^* (0, 4Mr^2]). This (t^*) *is the location of a local maximum of (|F|), and satisfies (|t^* - 0| =* t^{*} 4Mr2), i.e., (|T^* - | 4Mr^2) in the notation of Theorem 4.1. □

Remark 5.2.
The condition (M < 1/(4r)) ensures the perturbation is genuinely small relative to the pole strength on the interval ([-r, r]). For the application in Theorem 4.1, (M = O(N^{-} N)) as (N), so the condition is satisfied for all sufficiently large (N).

6. Arithmetic Control Experiments

To verify that the localization signal originates in genuine prime arithmetic rather than generic properties of oscillatory Dirichlet polynomials, we conduct three null-model experiments. These are computational in nature and are presented as supporting evidence, not as part of the analytic proof. The experiments are

implemented in the TEST_SUITE/ (referenced in the repository's README.md), specifically in TEST_NULL_MODELS.py.

6.1. Null Models

The three null models are constructed as follows.

(i) **Shuffled ((n))**: the multiset of values ({(n) : n N}) is randomly permuted across positions, preserving the magnitude distribution but randomizing prime locations.

(ii) **Random phases**: ((n)) is replaced by (|(n)| (i_n)), where (_n) are independent uniform random variables on ([0, 2]). This tests whether coherent arithmetic interference is required.

(iii) **Möbius ((n))**: ((n)) is replaced by the Möbius function ((n)), defined by ((n) = (-1)^k) if (n) is a product of (k) distinct primes, ((n) = 0) if (n) has a squared prime factor. Since (_{n} (n) n^{-s} = 1/(s)) (Davenport [1], Ch. 14), the Möbius kernel has poles at the zeros of ((s)), identical to the von Mangoldt kernel.

6.2. Results

In each case, (_N) is replaced by the corresponding modified kernel and zero-detection recall (fraction of known zeros for which a peak of the kernel lies within a fixed tolerance) is measured. The experiment uses (N = 2000) and (T), containing 10 known zeros (from RiemannZeros.txt).

Model	Recall	Relative to Baseline
Real ((n))	0.800	baseline
Shuffled ((n))	0.367	45.8%
Random phases	0.267	33.3%
Möbius ((n))	1.000	125.0%

Table 1. Null-model comparison at (N = 2000), (T). Recall = fraction of known zeros with a kernel peak within tolerance 0.3.

6.3. Interpretation

The shuffled ((n)) and random-phase models collapse to 46% and 33% of baseline respectively. This indicates that both the positions of primes and the coherent arithmetic interference structure of the sum are necessary for localization. Generic oscillatory Dirichlet polynomials of comparable magnitude do not replicate the signal.

The Möbius result (100% recall) is expected and provides structural confirmation rather than an anomaly. Since (1/(s) = (n) n^{-s}) has poles at the same zeros as (-'/(s)), the Möbius kernel is a second ()-linked arithmetic function independently detecting the same zero set. The agreement of two distinct arithmetic functions of this type, against the failure of random variants, supports the conclusion that the localization is genuinely arithmetic.

These experiments follow the methodology of arithmetic control tests in analytic number theory, analogous to the use of random Dirichlet polynomials as null distributions in zero-detection studies (cf. Montgomery [4], 1973, §7; Odlyzko [5], 1992, §3).

7. The Critical-Line Limit (Conjecture III(_))

The natural extension of Theorem 4.1 to the critical line (= 1/2) is the following open problem.

Conjecture 7.1 (Critical-Line Correspondence (III(_))).
The conclusions of Theorem 4.1 extend to (= 1/2) exactly: for a suitable scaling (N = N(T)), the local maxima of (|_N(1/2 + iT)|) are localized within (_N) of the zeros () of ((s)).

7.1. The Analytic Obstruction

The Davenport bound (3.2) degenerates as () in two simultaneous ways: the constant (C()), and the factor (N^{-}). Together, the right side of (3.2) no longer tends to zero, removing any control over (R_N) on the critical line.

To establish Conjecture 7.1 unconditionally would require one of the following: (a) a zero-free region of ((s)) adjacent to the critical line of positive width, whose existence is equivalent to a strong form of the Riemann Hypothesis (Titchmarsh [2], §3.8); (b) a subconvexity bound for ((s)) on the critical line sufficient to control partial sums of ((n) n^{-1/2-iT}) (Hardy and Littlewood [6], 1918; Weyl [7], 1921); or (c) a new analytic approach not currently available.

In each case the difficulty is of the same order as the Riemann Hypothesis itself. We regard Conjecture 7.1 as the central open problem of this programme.

7.2. Asymptotic Data

For reference, we record the behaviour of zero-detection recall when (_N) is evaluated at (= 1/2) (outside the theorem's domain) with (N(T) = C T T), the classical zero-density scaling. The declining recall as (T) increases is consistent with the absence of a rigorous bound at (= 1/2) and does not constitute evidence against Conjecture 7.1. These computations are implemented in TEST_CONJECTURE_III_LOCALIZATION.py.

(T)-window	(N(T))	Recall at (= 1/2) (illustrative only)
[10, 50]	574	0.900
[100, 200]	1381	0.800
[400, 800]	3589	0.650

Table 2. Asymptotic stability of recall at (= 1/2) with (N = C T T). These are computational observations outside the domain of Theorem 4.1.

8. A Scaling Note and Geodesic Motivation

The broader MKM programme uses a ()-scale parameter computed iteratively as the integer (k) satisfying (^k T), where (= (1 +)/2) is the golden ratio. This equals (_(T) = (T)/() (T)). Consequently (N(T) = C T)_scale((T) C T T), which is the classical zero-density scaling (Riemann [8], 1859; von Mangoldt [9], 1895). The () notation provides notational consistency within the broader framework but introduces no new mathematics beyond (T T) scaling. This is recorded here for transparency.

The geodesic motivation arises from embedding the arithmetic kernel into a 9-dimensional manifold where curvature singularities correspond to zeros. The intrinsic geodesic curvature is computed in GEODESIC_ARITHMETIC_ISOMORPHISM.py using only the von Mangoldt kernel and ()-weights, without learned parameters. An enhanced version with calibrated coefficients is provided for applications but is overfitted and not used for theorem claims.

9. Conclusions

We have established a conditional localization theorem for the smoothed von Mangoldt kernel (_N(s)) off the critical line. Assuming standard remainder and derivative bounds (H1, H2) for exponentially

smoothed Dirichlet series, the local maxima of (|_N(1/2 + + iT)|) are localized within (_N() = O(N^{-} N)) of the Riemann zeros, with (_N()) as (N) for any fixed (> 0).

Three null-model experiments confirm that this localization is arithmetic in origin: randomizing prime positions or phases reduces recall by 54-67%, while an independent ()-linked arithmetic function (Möbius ()) achieves full recall, consistent with the theoretical prediction.

The critical-line limit (Conjecture 7.1) is precisely identified as the () boundary where the remainder bound degenerates. Resolving this conjecture requires analytic control of (-'/) on ((s) = 1/2) that is not currently available.

The two main items of future work are: (a) filling in the explicit constants in H1 and H2 to obtain a fully self-contained analytic proof; and (b) identifying a new mechanism for the critical-line case.

References

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[S1] Mullings, J. Supplementary validation script and data for "The Riemann Hypothesis: The Singularity Proof - Conjecture III"
https://github.com/jmullings/riemann-hypothesis-singularity-proof/tree/main/CONJECTURE_III .
Includes all code for kernel definitions, bounds, lemmas, verifications, and null models.